DINAL UDAGEDARA

Software Engineer

BEng (Hons) Software Engineering - University of Westminster

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PROFILE

Results-driven Software Engineer with approximately 2 years of professional experience, including an 11-month industrial placement and promotion to a full-time role. Holds a BEng (Hons) in Software Engineering (First Class Honours) from the University of Westminster. Specializes in full-stack development, Web3 integration, and AI-driven platforms. Experienced in building and integrating smart contract-backed systems, developing scalable RESTful APIs, and leading end-to-end feature delivery. Previously engaged as the primary software engineer for a US-based AI music generation and streaming startup.

TECHNICAL SKILLS

Programming Languages: JavaScript, TypeScript, Python, Java

Frontend: React, Next.js, HTML, CSS

Backend: Node.js, Express.js, NestJS, FastAPI

Database: MongoDB, PostgreSQL, SQL

Web3 / Blockchain: Smart Contract Integration, Web3.js / Ethers.js

Cloud & DevOps: AWS (Amplify, CloudFront, EC2, S3 - working knowledge), Docker (working knowledge), Git, GitHub

AI / LLM: Prompt Engineering, Structured Outputs / Constrained Decoding, RAG Systems (LangChain, OpenAI Embeddings), AI Agent Design: multi-model workflows across OpenAI and Gemini in production systems

Dev Tools: Postman, VS Code, Figma, Cursor, Claude, OpenAI Codex

WORK EXPERIENCE

Software Engineer • Perfectus Technologies *Feb 2025 – Present*

Perfectus

- Develop and maintain full-stack web applications using modern frameworks across both frontend and backend layers, including resolving a persistent AWS Amplify / CloudFront DNS conflict that unblocked a stalled client production deployment.
- Integrate smart contracts into web platforms, connecting decentralized on-chain logic with traditional Web2 backends and frontends (Web3 integration without smart contract authoring).
- Architect and implement RESTful APIs and backend services to support scalable, production-grade applications, including diagnosing and resolving a live production database outage that restored platform availability with minimal downtime.
- Collaborate in an agile team environment, participating in code reviews, sprint planning, and feature delivery — including consolidating services under a unified domain to resolve corporate firewall whitelisting issues that were blocking an enterprise client integration.
- Promoted to Software Engineer after a successful 11-month internship, reflecting strong technical contributions and growth.
- **Overall contribution:** Consistent feature delivery, infrastructure fixes, and code quality across multiple client projects through both the internship and full-time role, earning a promotion to Software Engineer.

Software Engineering Intern • Perfectus Technologies *Mar 2024 – Jan 2025 (11 months)*

- Completed an 11-month industrial placement as part of the BEng (Hons) Software Engineering programme at IIT.
- Contributed to full-stack development across multiple client projects, gaining hands-on experience in production environments.
- Worked on Web3-enabled web applications involving smart contract integration on the frontend and backend.
- Collaborated with senior engineers to deliver features, fix bugs, and improve code quality across the codebase.

Freelance Software Engineer • Independent - US Client (Remote) *2025 – August 2026*

- Served as the primary software engineer for a US-based AI music generation and streaming platform.
- Led end-to-end feature development across the full stack, from backend APIs to frontend user interfaces.
- Worked directly with the client to define technical requirements, architect solutions, and deliver production-ready code.
- Built and integrated AI-powered music generation capabilities into a scalable streaming platform.
- **Overall contribution:** As the sole engineer on the platform, helped the startup go from concept to a shipped, scalable product across roughly a year and a half of active development.

EDUCATION

Informatics Institute of Technology (University of Westminster, UK)

BEng (Hons) Software Engineering with Industrial Placement, First Class Honours

2023 – 2026 | Awarded 12 June 2026

Includes 11-month industrial placement at Perfectus Technologies.

St. Joseph Vaz College, Wennapuwa

Advanced Level - Physical Science Stream | Completed 2022

- Completed A/L examinations in the Physical Science Stream
- Served as a School Prefect

PROJECTS

CrackInt – AI-Driven Personalized Interview Preparation Platform *2025 – 2026*

Final Year Project | *crackint.dinaludagedara.com* | *Source (Frontend)* | *Source (Backend)*

Tech Stack: Next.js, FastAPI, PostgreSQL, Word2Vec, BiLSTM-CRF

- Full-stack AI interview preparation platform built with Next.js and FastAPI, backed by PostgreSQL with modular API routing.
- Designed and trained a Word2Vec + BiLSTM + CRF NLP pipeline for Named Entity Recognition (NER) over résumé and job-posting text, achieving a test micro-F1 of 0.83 on 4,738 annotated résumés and ~0.85 on 6,327 job postings.
- Platform features AI tutor chat, targeted interview question practice, skill gap analysis, readiness scoring, and tailored cover letter generation.
- LLM-enabled features are controlled through configuration flags for reliable deployment flexibility.
- Fully deployed and live at crackint.dinaludagedara.com.

SAPTalk – Natural-Language Query Interface for SAP Business Data *2026*

saptalk.dinaludagedara.com | *Source Code*

Tech Stack: NestJS, Next.js 16, TypeScript, Zod, OpenAI Structured Outputs, SAP OData V2

- Built a natural-language interface over the SAP S/4HANA Business Partner API (SAP Business Accelerator Hub sandbox) in which the language model never writes a query - it emits a structured intent that is validated against a field allowlist before deterministic code compiles it to OData.
- Generated the model's JSON Schema from a single field registry so hallucinated fields cannot be emitted, with server-side validation of operator legality and value formats that JSON Schema cannot express; rejected intents are retried with the validator's errors.
- Escaped all model-supplied values at the compiler boundary, so an injected OData clause compiles to an inert string literal - verified against the live sandbox API.
- Implemented cross-entity questions as two joined requests after confirming OData V2 cannot filter across navigation properties, choosing join order by selectivity to keep results exact rather than truncated.

Showdown – No-Limit Texas Hold’em Poker Platform *2026*
poker.dinaludagedara.com | *Source Code*
Tech Stack: Next.js 16, React 19, TypeScript, Redis, Server-Sent Events, Tailwind CSS

- Built a full no-limit Texas Hold'em table with Redis-backed real-time multiplayer over server-sent events (SSE), supporting public and private rooms for 2–6 seats.
- Designed the poker engine as a pure, deterministic state machine (hand ranking, side pots, escalating blinds, turn clock) fully decoupled from HTTP and storage, so multiplayer was added without changing any core game logic.
- Implemented per-viewer hole card redaction on the server so the browser never receives a card a player isn't entitled to see.
- Built equity-aware bots that open with a Chen-formula preflop range and switch to Monte Carlo equity simulation from the flop onward, with bot fill for empty seats and rematch into a fresh room when a table finishes.

Sri Grow – Agricultural Advisory Platform for Rural Sri Lanka (Group) *2024*
sri-grow.vercel.app | *Source Code*
Tech Stack: React, Node.js/Express, Python Flask, Random Forest Regressor

- Full-stack agricultural advisory web app targeting rural agricultural officers in Sri Lanka, supporting crop planning, pest alerts, and market price tracking.
- Trained a Random Forest Regressor on a custom Sri Lanka weather dataset to predict maximum precipitation by city and date range, enabling data-driven crop scheduling.
- Served the ML model as a Python Flask microservice alongside a Node.js/Express backend and React frontend - a 3-tier architecture with separate ML inference layer.
- Features include live weather forecasting, crop-specific soil and weather guidance, real-time pest alerts, and current market prices sourced from structured datasets.

srt-lyric-player – Spotify-Style Lyrics Player & npm Package *2025*
srt-to-lyrics-ebon.vercel.app | *npm* | *Source Code*
Tech Stack: Next.js 15, TypeScript, Howler.js, Framer Motion, Web Audio API, tsup

- Built a Spotify-style music player with real-time synchronized lyrics driven by standard .srt subtitle files, and published the result as the srt-lyric-player npm package.
- Engineered real-time lyric sync using Howler.js for audio playback, Framer Motion for lyric transition animations, and Web Audio API for a canvas-based audio visualizer.
- Structured as a Next.js 15 monorepo with the package bundled via tsup (ESM + CJS + TypeScript types) and deployed live on Vercel as an interactive demo.

CERTIFICATES & ACHIEVEMENTS

- Benchmark 2.0 UI/UX | TOP 10 Finalist
- Completed British Council's COL Skills Plus 2 Certification
- Completed ESOFT Diploma in Information Technology (DiTEC)
- Blogs about my software engineering journey on Medium (medium.com/@dinal.bandara)

EXTRA CURRICULAR ACTIVITIES

- IEEE SIGHT Volunteering
- Represented School Chess Team
- Represented School Badminton Team

REFERENCES

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